

Anotation

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In current educational policy, the development of computational thinking is regarded as a key competency, the systematic assessment of which requires valid and reliable diagnostic tools. This paper presents the results of an extensive study focused on the construction, psychometric validation, and standardization of a computational thinking test designed for fifth-grade elementary school students in the Czech Republic.

Based on a systematic analysis of international theoretical frameworks and an empirical survey of teachers, an operational definition of computational thinking was developed, comprising three latent dimensions: Algorithmic Thinking, Abstraction and Debugging, and Syntax and Coding. Based on this definition, a 12-item test was constructed, which was pilot-tested and subsequently administered to a sample of 741 students.

Psychometric analyses confirmed the instrument's high internal consistency ($\alpha = 0.813$) and adequate construct validity. The study also includes the creation of percentile norms and a proposed classification scale enabling the interpretation of individual results. Analyses further showed that the level of computational thinking is not statistically significantly influenced by the students' gender or age, with the highest performance recorded in the Syntax and Coding dimension.

This entry offers not only a new standardized tool for diagnosing computational thinking but also a detailed demonstration of methodological procedures relevant to the construction and validation of educational assessment tools.